MULTI-WALLED CARBON NANOTUBE

Characterization Analysis Report

Report generated: 12 September 2026

This report presents the analysis of the available multi-walled carbon nanotube characterization data. The analysis uses the processed experimental datasets available in the project data directory.

1. Characterization Summary

Parameter	Result
BET surface area (m²/g)	57.239
BJH adsorption mode radius (nm)	21.6487
BJH adsorption mode diameter (nm)	43.2974
BJH adsorption pore volume (cm³/g)	0.409953
BJH desorption mode radius (nm)	12.7284
BJH desorption mode diameter (nm)	25.4568
BJH desorption pore volume (cm³/g)	0.418254
DFT peak pore width (nm)	22.127
DFT pore volume (cm³/g)	0.364711
DFT surface area (m²/g)	66.5704
XRD strongest intensity point (2θ)	31.3156
XRD strongest intensity	1807.0
XRD detected peaks	516
EDS element count	9
EDS total weight (%)	99.99999999999999
SEM image count	10
TEM image count	3

2. General Evaluation

• BET analysis provides a measurable specific surface area.

• BJH adsorption data provide a pore-size distribution.

• BJH desorption data provide a pore-size distribution.

• DFT analysis provides a pore-width distribution.

• EDS provides elemental composition data.

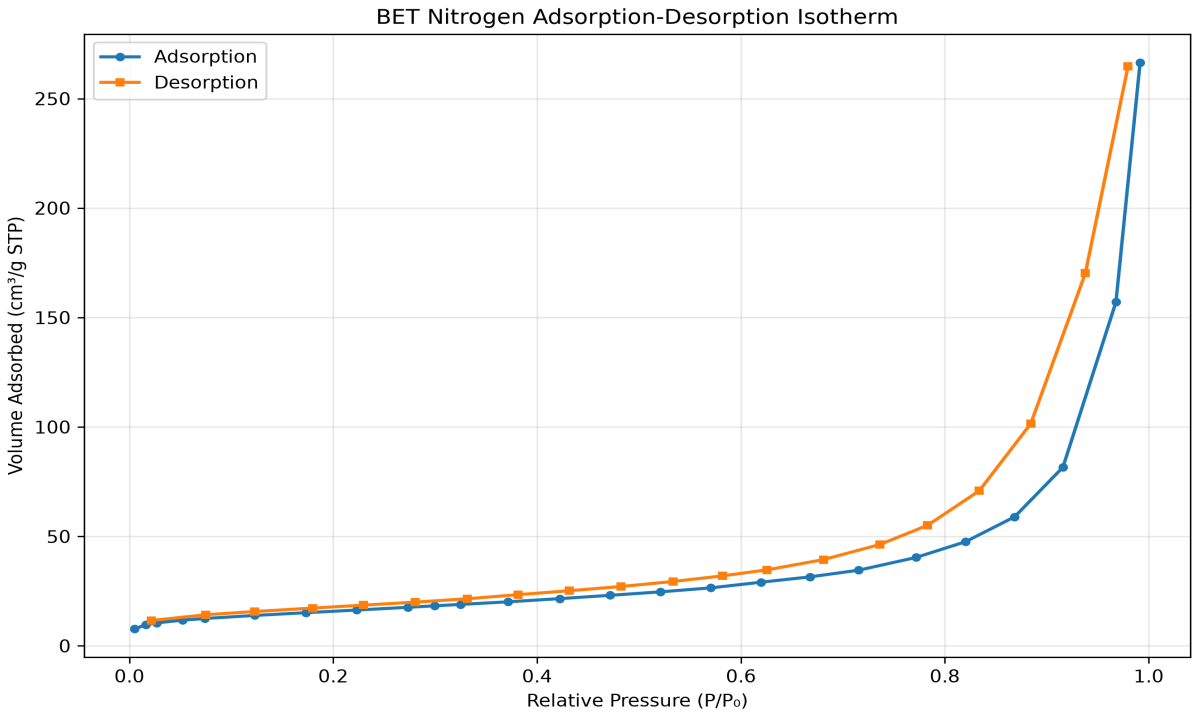
• XRD data contain detected local intensity maxima.

- 10 SEM images are available for morphological examination.
- 3 TEM images are available for morphological examination.

3. BET Surface Area and Isotherm Analysis

The extracted BET dataset reports a specific surface area of 57.2390 m²/g. The adsorption data were used to generate the available isotherm plot.

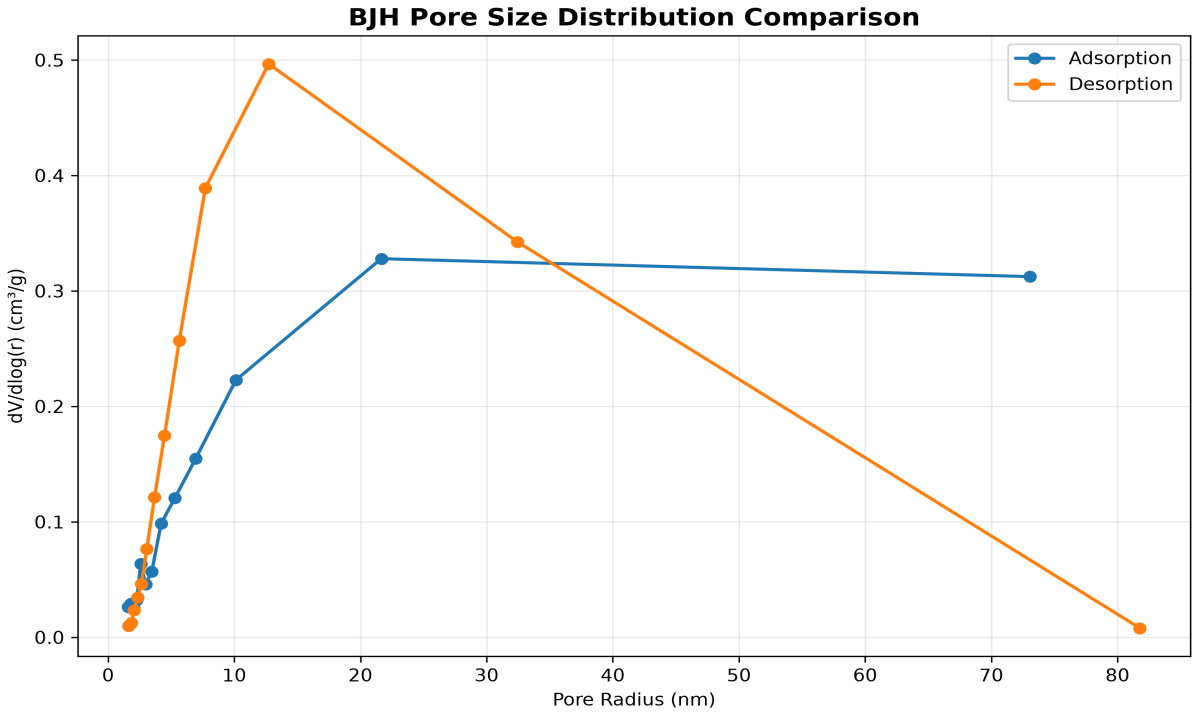
BET Adsorption-Desorption Isotherm



4. BJH Pore Size Distribution

The calculated mode of the BJH adsorption pore radius distribution is 21.6487 nm. The maximum cumulative pore volume represented in the extracted adsorption dataset is 0.4100 cm³/g.

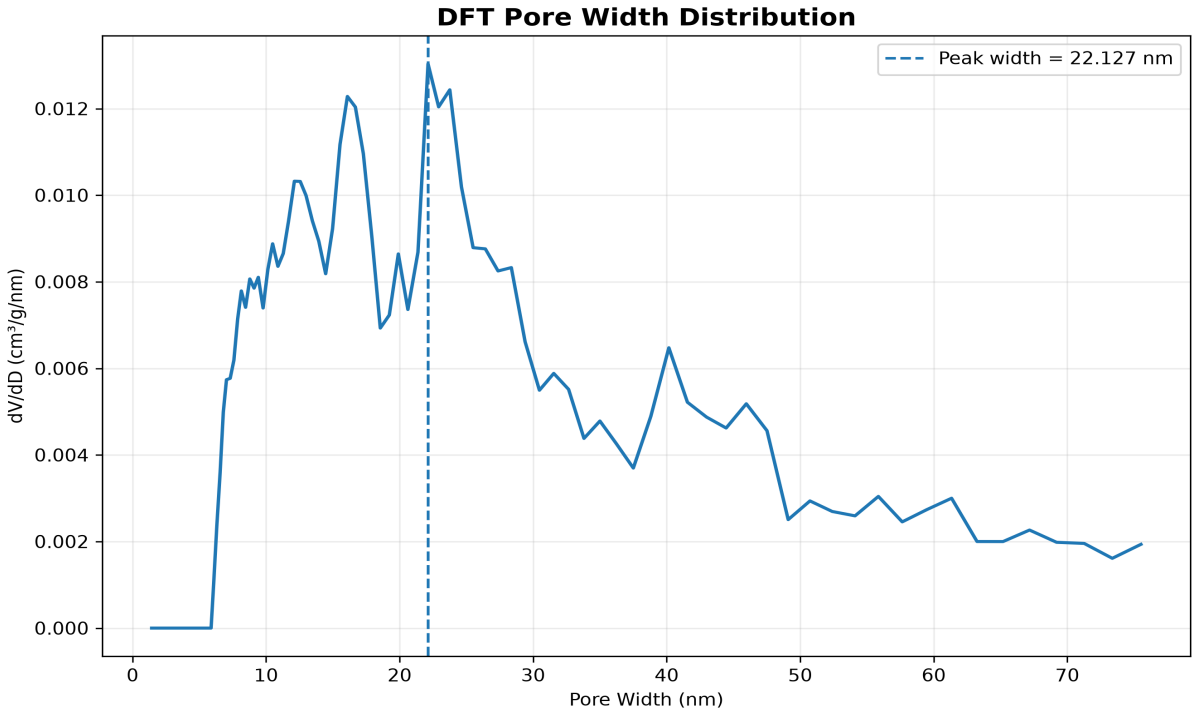
BJH Adsorption and Desorption Comparison



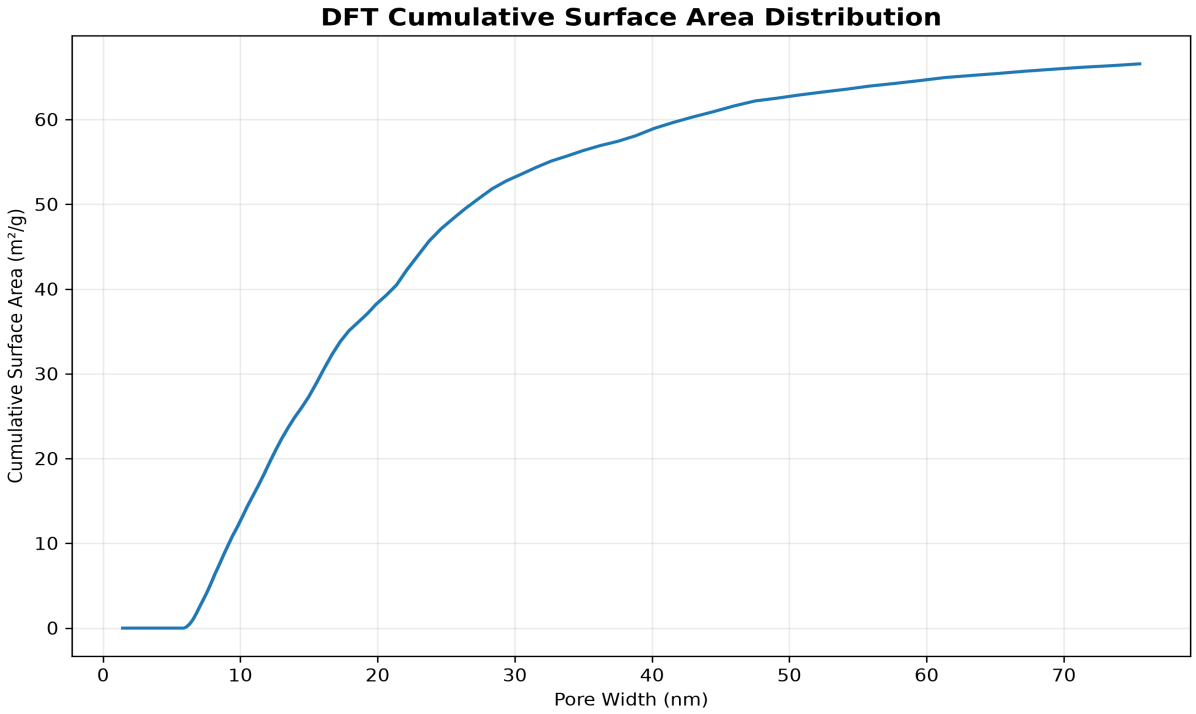
5. DFT Pore Structure Analysis

The DFT pore-size dataset indicates a maximum differential pore-volume response at approximately 22.1270 nm within the available dataset.

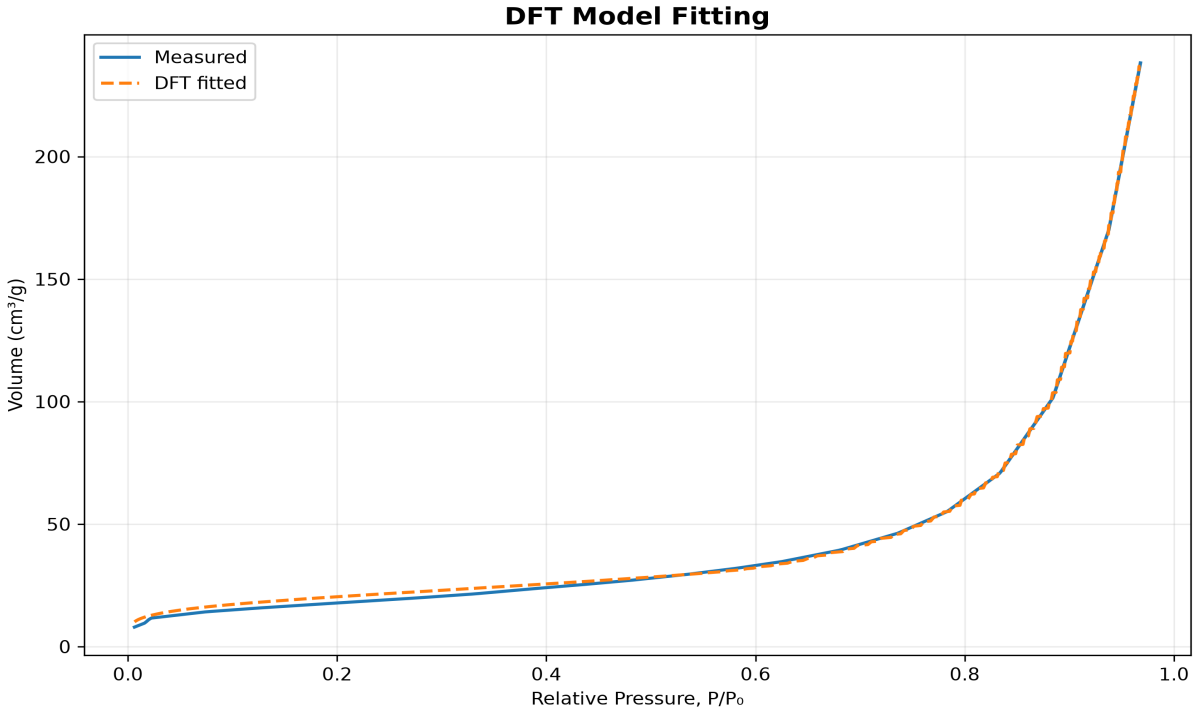
DFT Pore Size Distribution



DFT Surface Area Distribution



DFT Model Fitting

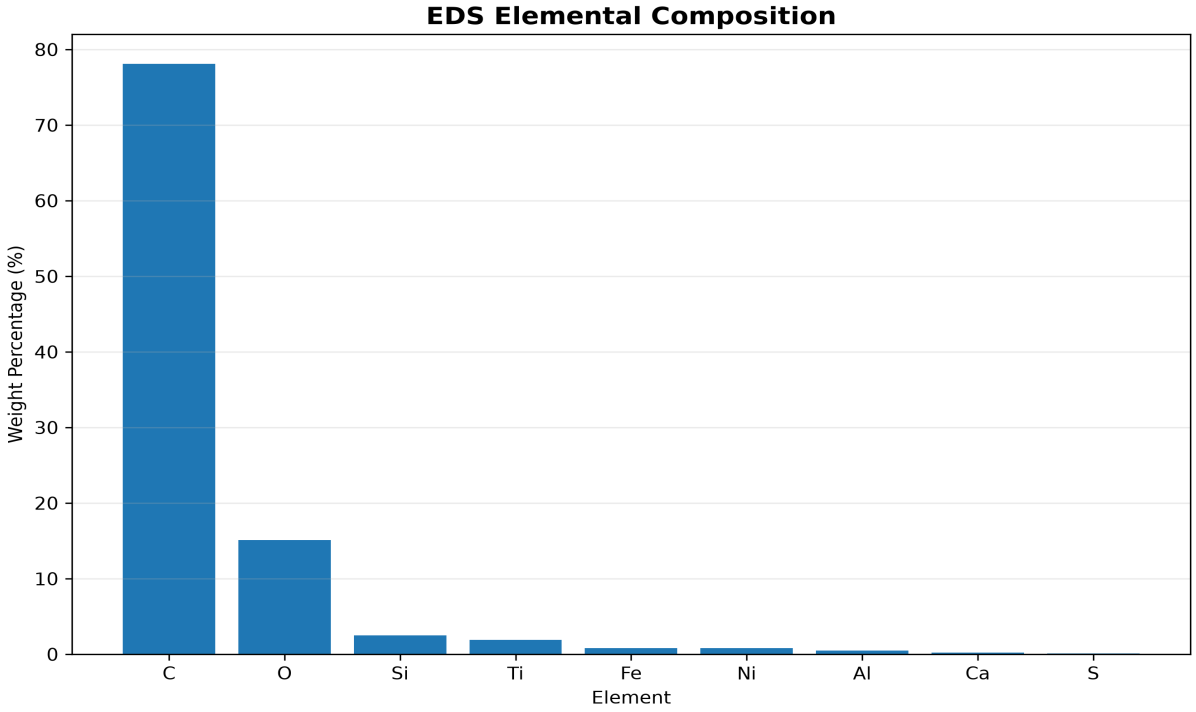


6. EDS Elemental Composition

The EDS dataset contains 9 reported elements. The sum of the reported weight percentages is 100.00%.

Element	Weight (%)	Sigma
C	78.10	0.40
O	15.10	0.40
Si	2.50	0.00
Ti	1.90	0.10
Fe	0.80	0.10
Ni	0.80	0.10
Al	0.50	0.00
Ca	0.20	0.00
S	0.10	0.00

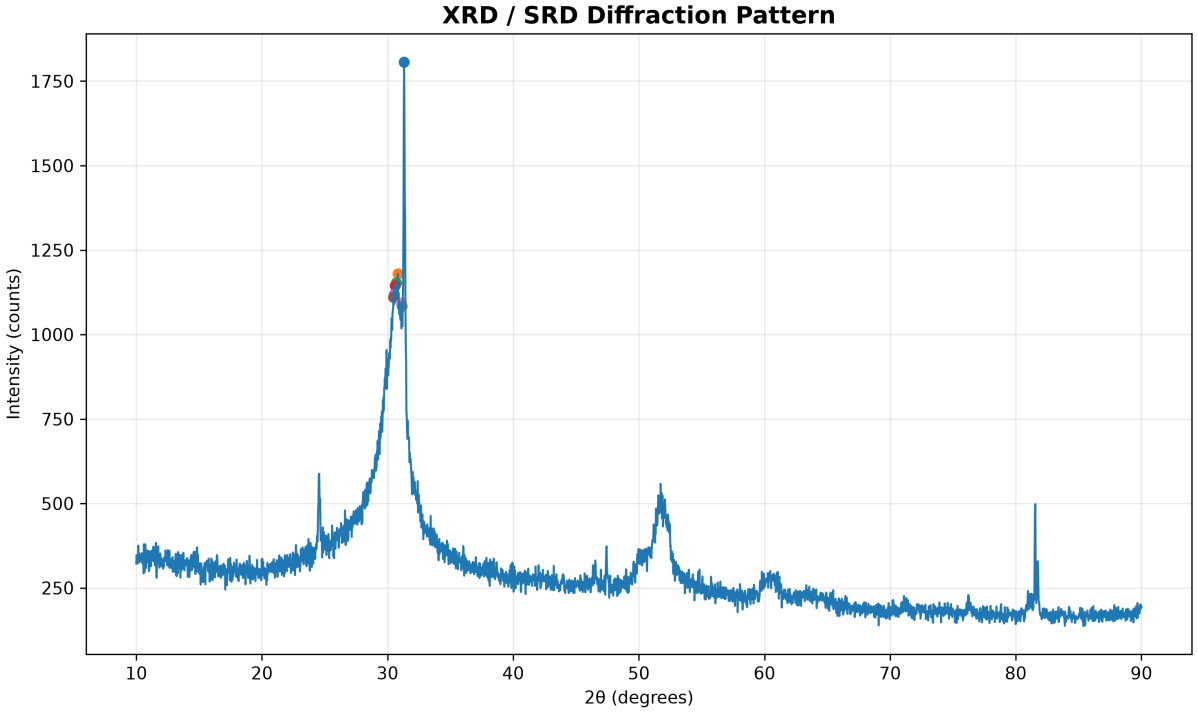
EDS Elemental Composition



7. X-Ray Diffraction Analysis

The highest intensity point in the supplied XRD pattern occurs at approximately 31.3156° 2θ, with an intensity of 1807 counts.

X-Ray Diffraction Pattern



8. SEM and Morphological Data

The project contains 10 SEM image records with associated metadata. The available images can be examined separately for morphological interpretation. Quantitative dimensions should only be reported when they are directly measured from the image scale.

9. Conclusion

The characterization dataset provides complementary information concerning the surface area, pore structure, elemental composition, crystal-related diffraction features, and available morphology data of the MWCNT sample. BET, BJH and DFT analyses provide surface and pore-structure information, while EDS and XRD provide compositional and structural information. SEM data provide the basis for morphological examination.

The numerical values presented in this report are derived from the processed datasets available to the application. Further scientific interpretation should be made alongside the original experimental conditions and instrument reports.